

1. Executive Summary & Architecture Topology

Fizmoh is an enterprise multi-tenant WhatsApp Commerce, Operational Automation, and Conversational AI SaaS platform. It enables specialized vertical businesses to operate end-to-end customer commerce and support over WhatsApp, while providing platform administrators with a centralized control plane.

Platform Operator Console

Endpoint: /platform/*

- Command Center & Global KPI Analytics
- Multi-Tenant Lifecycle & Impersonation
- SaaS Subscriptions, Invoicing & Dunning
- Meta WhatsApp Number Fleet Management
- Resource Quotas, Support & Audit Logs

Tenant Workspace SaaS Surface

Endpoint: /:tenant-slug/*

- 2-Way Omnichannel WhatsApp Live Inbox
- AI Assistant Grounded in Knowledge Base
- Tour Bookings, Hospital & Restaurant POS
- Broadcast Campaigns & Meta Templates
- AmwalPay Gateway & WooCommerce Sync

Core Technical Specifications

Architecture Layer	Technology / Library	Architectural Role & Function
Frontend Framework	Next.js 16 (Turbopack)	App Router, React 19, Server Components
Backend & APIs	Next.js Edge / Node APIs	Dynamic Route Handlers & Middlewares
Database & ORM	PostgreSQL + Prisma 6.x	Multi-tenant relational persistence & indexes
UI & Visualization	Tailwind CSS + Recharts	Radix UI / shadcn, Lucide, Interactive Charts
Integrations	Meta Cloud API, AmwalPay	WABA Webhooks, Google Meet, WooCommerce

Security & Isolation Highlights

- Complete Multi-Tenant Data Isolation enforced at database query & API middleware levels.
- Zero-Knowledge Operator Impersonation with mandatory audit logging and reason capture.
- Type-Safe Endpoints & Dynamic Routing without legacy URL hash fragments.
- Automated Dunning Engine with payment gateway webhook reconciliation.
- Immutable Audit Trail logging all administrative actions with 1-click CSV export.

2. Platform Operator Console Features (/platform/*)

The Platform Operator Console is a standalone command surface built for Fizmoh global administrators. Every section is accessible via clean, direct URL paths with full-width responsive layouts and CRUD controls.

Command Center & Tenant Workspace Governance

Routes: /platform/command-center, /platform/tenants

- Real-time Platform KPIs: Active business counts, platform MRR, WhatsApp messages, orders & leads.
- Interactive Trend Visualizations: Recharts area & bar charts illustrating message & revenue velocity.
- Tenant Lifecycle Management: Provision new workspaces, suspend delinquent accounts, resume services.
- Impersonation Gateway: Open and inspect tenant workspace sessions with audit trail tracking.
- Unrouted Number Detector: Captures inbound messages from unregistered WhatsApp numbers.
- Custom Domain Configuration: Bind dedicated domains (e.g., booking.agency.com) to workspaces.

Subscriptions, Invoicing & Automated Dunning Engine

Routes: /platform/billing, /platform/invoices, /platform/revenue

- Plan Assignment & Modification: Switch tenant plans (Starter, Growth, Enterprise) with instant quota adjustments.
- 1-Click Trial Extensions: Extend trial periods by configurable day increments with automatic date recalculation.
- Automated Dunning Escalation: Multi-step dunning reminders (1st Reminder -> 2nd Notice -> Final Warning).
- Invoice Lifecycle: Track OMR amounts, issue dates, payment gateway IDs, and 1-click 'Mark as Paid' actions.
- Financial Intelligence: Live calculation of MRR, ARPU (Average Revenue Per User), and 30-day collections.
- Status Badges: Clear visual indicators for ACTIVE, TRIALING, PAST_DUE, CANCELLED subscriptions.

WhatsApp Number Pool & Resource Quotas

Routes: /platform/whatsapp-numbers, /platform/usage

- Meta Cloud API Health Monitor: Real-time quality rating indicators (Green Normal, Yellow Warning, Red Restricted).
- Messaging Tier Allocation: Displays tier throughput limits (Tier 1K, Tier 10K, Tier 100K, Unlimited/day).
- Number Pool Allocation: Assign phone numbers to dedicated workspaces or hold in unassigned pool.
- Resource Quota Matrix: Visual progress bar monitoring per workspace for messages, campaigns, AI tokens & staff.
- High-Usage Detection: Workspaces exceeding 90% quota are automatically highlighted with alert badges.
- Manual Re-Sync & Diagnostics: Trigger instant health checks against Meta Graph API endpoints.